

Progettazione di Applicazioni Mobili

Mobile app design

Envision – Part 2/2

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Storyboarding

- A storyboard is a sequence of visual “frames” illustrating the interplay between a user and an envisioned system
- Storyboards bring the design to life in graphical “clips”, freeze-frame sketches of stories of how people will work with the system
- A storyboard might be thought of as a “comic-book” style illustration of a scenario, with actors, screens, interaction, and dialogue showing sequences of flow from frame to frame

Example

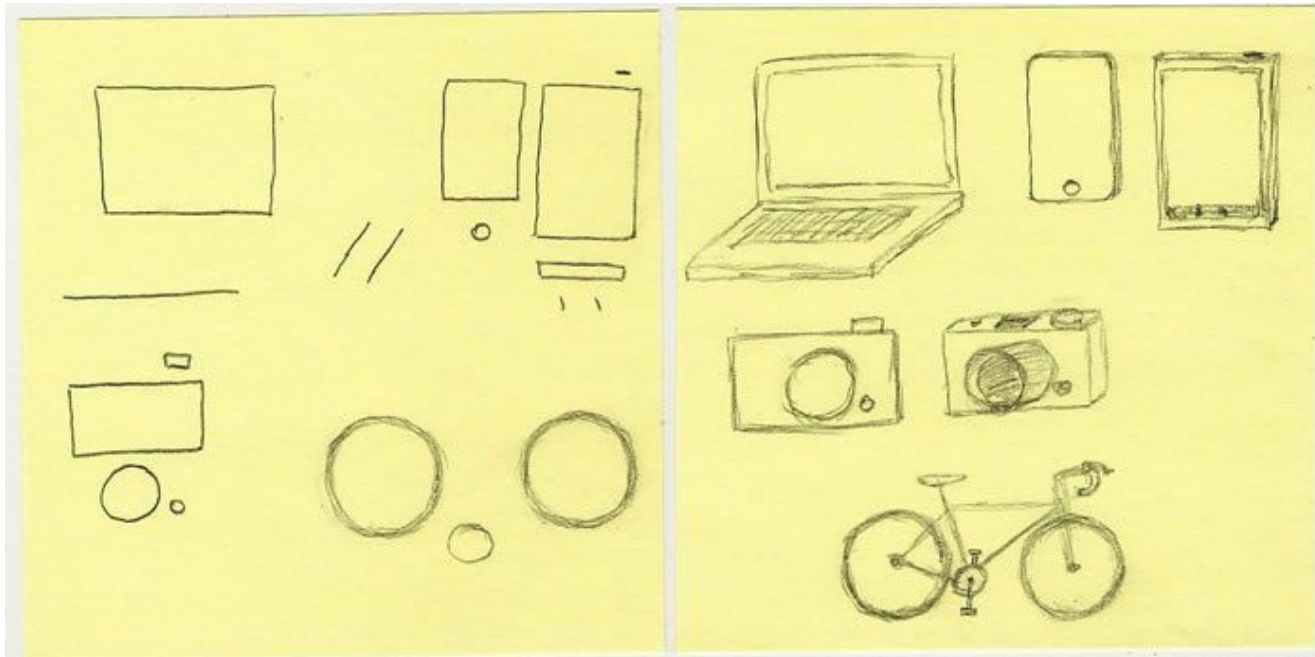


Creating a storyboard

- Use small square sticky notes (post-it) and a pencil
- Types of elements typically included in most frames in a storyboard:
 - objects
 - people
 - emotions (through facial expressions)
 - dialogues

Objects

- Objects can be created by combining simple shapes
- Objects help add realism and understanding to storyboards



- Accurate drawing of objects is more important to creating story realism than accurate drawings of people
- However, avoid giving too much detail
- Do not draw full interfaces!
 - Show no more than three user interface elements
 - Minimize the number of frames including an interface

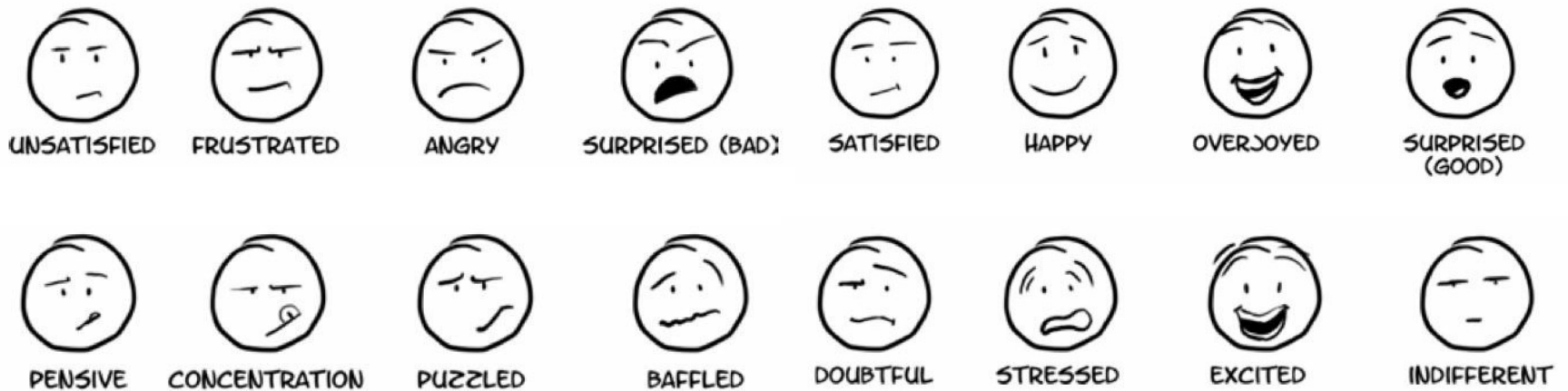
People

- Drawing people accurately is harder than drawing objects but simple stick figures are enough to communicate a vision



Emotions

- The facial expression recognition engine of most humans is even more sophisticated than figure recognition
- Just drawing two dots and a line allows people to see a face with multiple facial expressions



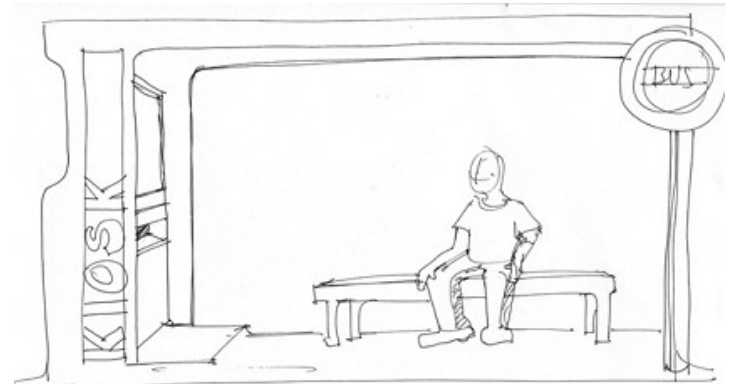
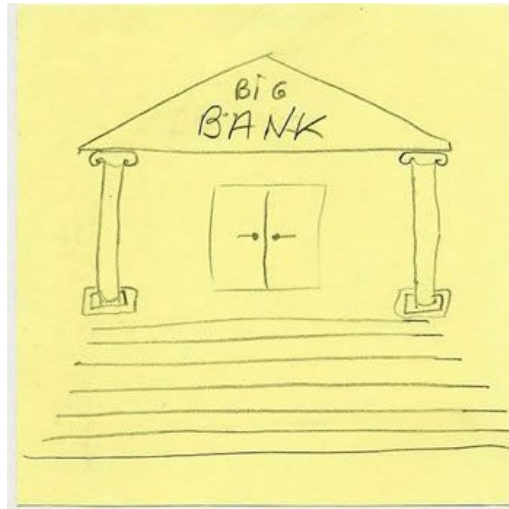
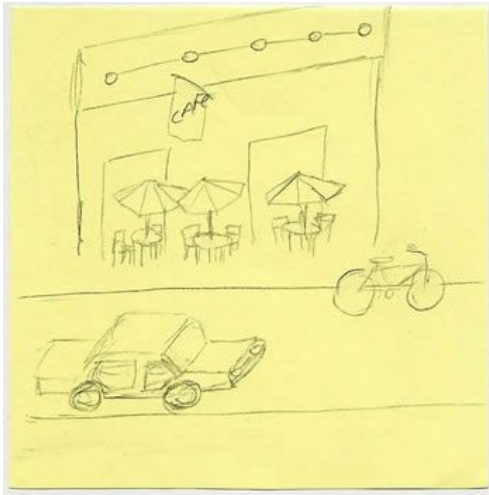
Dialogues

- Avoid captions, use dialogue and thought balloons



The establishing shot

- The establishing shot is the (first) frame of a storyboard and answers the question “Where am I?”
- The more you can set up the atmosphere and provide the setting for the story in the establishing shot, the less you have to work at it in the remainder of the storyboard



Transitions

- Pick appropriate transitions



1. MOMENT TO MOMENT



A SINGLE ACTION PORTRAYED IN A SERIES OF MOMENTS.



2. ACTION TO ACTION



A SINGLE SUBJECT (PERSON, OBJECT, ETC...) IN A SERIES OF ACTIONS.



3. SUBJECT TO SUBJECT



A SERIES OF CHANGING SUBJECTS WITHIN A SINGLE SCENE.



4. SCENE TO SCENE



TRANSITIONS ACROSS SIGNIFICANT DISTANCES OF TIME AND/OR SPACE.



5. ASPECT TO ASPECT



TRANSITIONS FROM ONE ASPECT OF A PLACE, IDEA OR MOOD TO ANOTHER.



6. NON SEQUITUR



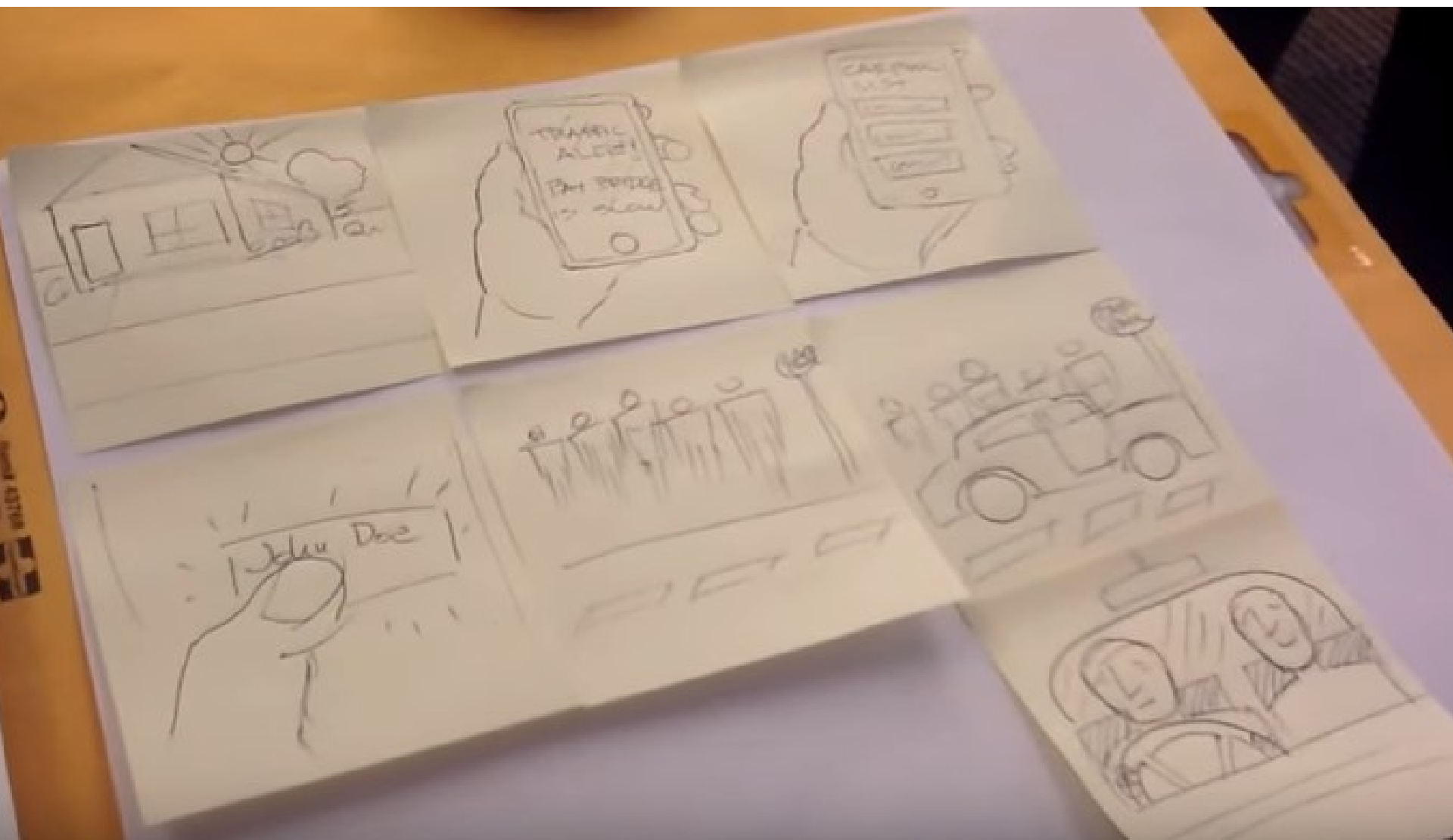
A SERIES OF SEEMINGLY NONSENSICAL, UNRELATED IMAGES AND/OR WORDS.

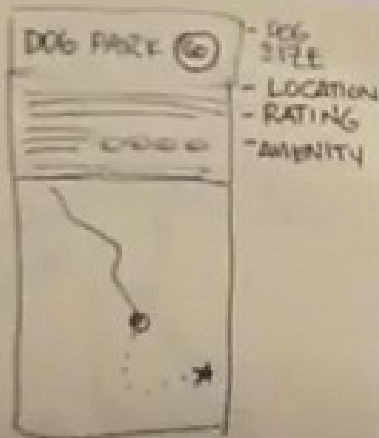
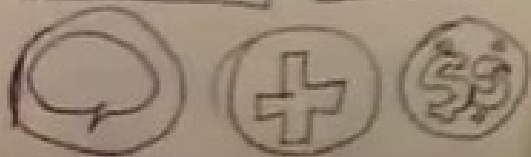
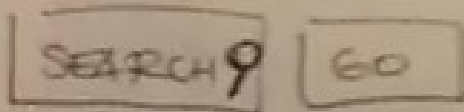
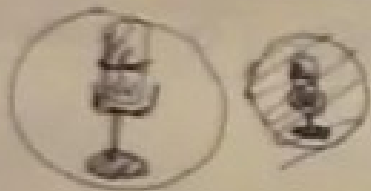
General guidelines

- Stick to five to eight frames total
- Start by asking the question: “Where am I?”
- Spend just enough time on your storyboard to document your vision and then stop
- Do not add too much interface detail
- Close with an (appropriate) bang
 - the final frame of the storyboard should show the benefit of using the app
 - stay believable

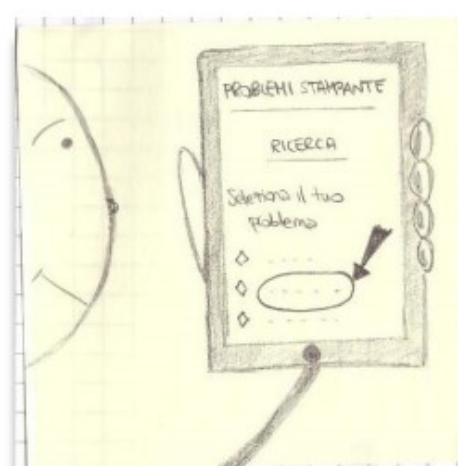
Testing the storyboard

- After you complete a storyboard, test it!
- You should first do a walkthrough with your team to understand if the storyboard is appropriate
 - The walkthrough may last only 10-15 minutes
 - Walk through each frame and record feedback from the team (is everything clear? are there missing elements? ...)
 - Modify the storyboard if needed
- Test the storyboard with your prospective users to check if they understand and relate to it!









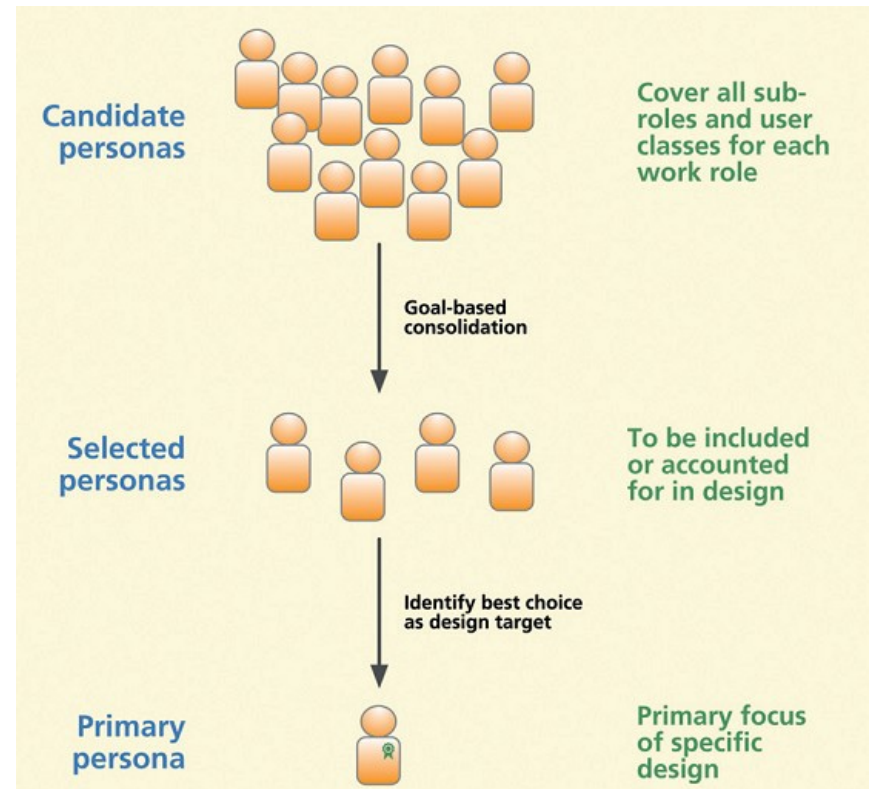
User personas

- A design for a broad user population should have the richest possible range of functionality, with maximum flexibility in how users can pick the parts they like the most, right?
- Wrong! You simply cannot make a single design be the best for everyone
- It is better to have a small percentage of the user population completely satisfied than the whole population half satisfied
- The logical extreme of this idea is to design for one user. This is where personas come in.

- A persona is a “hypothetical archetype”, a story and description of a specific individual who has a name, a life, and a personality
- A persona is based on real users but it is not an abstract user with needs and characteristics averaged across many other kinds of users
- The description of a persona needs to make it so well defined as a real and living being that it is impossible for a designer or programmer to substitute themselves or their own characteristics when creating the design

Creating personas

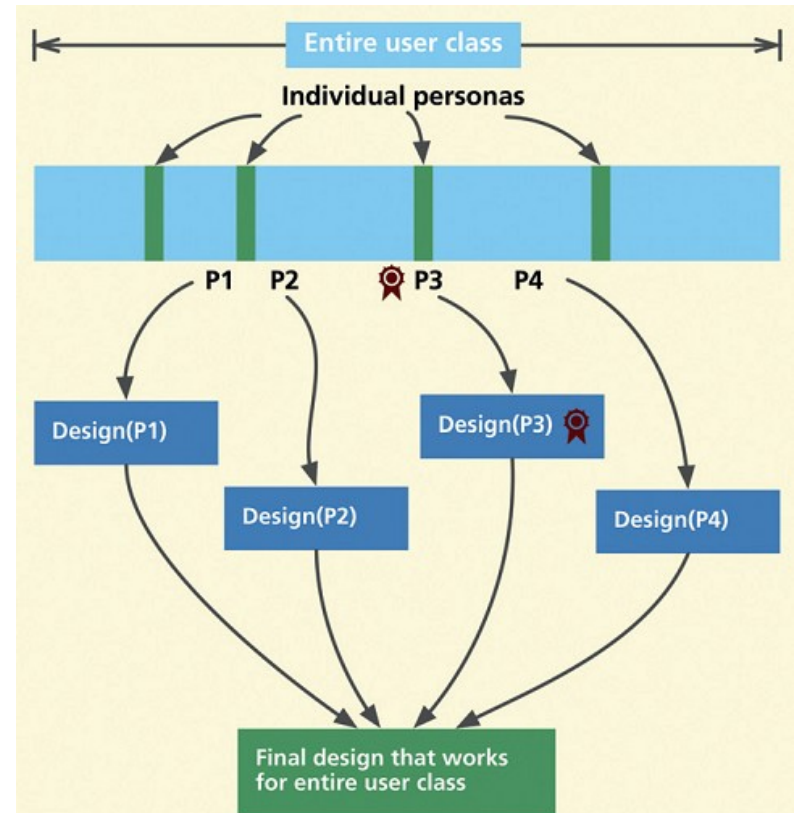
- For any given work role, personas are defined by user goals arising from their sub-roles and user classes. Different sub-roles and associated user classes have different goals, which will lead to different designs.
- It is typical to create multiple candidate personas generally corresponding to a major sub-role or user class. Usually, candidate personas are identified on the fly while interviewing potential users.
- How many candidate personas do you need? As many as it takes to cover all the users. It could be in the dozens.



- Your persona should have a (fictitious) name to make it personal and real
- Possibly, mockup a (fictitious) photo of this person
- Write some short textual narratives about their work role, goals, main tasks, usage stories, problems encountered in work practice, concerns, biggest barriers to their work, etc.
- The detail of a persona has to be a rich part of a life story. It has to be specific and precise; this means lots of details that all fit together. Give your persona a personality and a life surrounded with detailed artifacts.

- The next step is to merge personas that have similar goals. Merging reduces the number of personas that you must consider but you still cannot design for a whole group of personas that you may have selected.
- Choose one of the personas selected in the previous step as the one primary persona, the single best design target, the persona to which the design will be made specific
- The design specifically for the right primary persona will at least work for the others, but a design specifically for any of the other selected personas may not work for the primary persona

- The idea behind designing for a persona is that the design must make the primary persona very happy, while not making any of the selected personas unhappy
- As you converge on the final design, the nonprimary personas will be accounted for, but will defer to this primary persona design concerns in case of conflict
- If there is a design trade-off, you will resolve the trade-off to benefit the primary persona and still make it work for the other selected personas



Karla



"I'm always looking for ways to destress but group classes don't work with my schedule."

Age: 45

Role: Physician

Location: Chicago, IL

Education: M.D.

Family: Married w/ 2 kids

Motivators



Technology



Brands



Whole Foods



Facebook



Starbucks

About

Karla is a primary care physician in a busy hospital. She values her family time and doesn't like when stress from work gets in the way. The hours she works make it hard to get to group meditation classes, although she knows how important it is for her health.

Personality



Frustrations

- Hard-to-learn interfaces
- No variety of guided meditations
- Subscription with too many choices, not enough guidance

Goals

- Learn new meditation methods
- Start self-care a routine for stress
- Get better sleep

USER PERSONA



Lisa Mason

"I love learning about new cultures when I travel, it's my favorite part!"



AGE	35	STATUS	Single
JOB TITLE	Project Coord.	LOCATION	San Diego, CA

ABOUT

Lisa enjoys traveling domestically as well as internationally. She travels 1-4 times per year, depending on how much vacation time she has available. She's not a big fan of planning trips, but she likes finding fun things to do in new places. She's a big foodie and loves finding secret, local favorite restaurants. She also enjoys interacting with local people and learning about their culture.

GOALS

- Focus on local cultural activities instead of typical tourist activities
- Learn local phrases before a trip to communicate better with the locals
- Learn about the local customs ahead of time

KNOWN HABITS

- Use Google Maps and Yelp to research upcoming trips.
- Takes her friends recommendations on places to go that she's never been to.
- Going to local hangouts when she travels, (restaurants, bars, clubs) to connect with the local lifestyle.

PREFERENCES

- | | |
|-----------------------|-----------------------|
| Local Food | <input type="range"/> |
| Learn New Cultures | <input type="range"/> |
| Local Activities | <input type="range"/> |
| Seeing New Places | <input type="range"/> |
| Tropical Destinations | <input type="range"/> |

PAIN POINTS

- Communicating with local people when there is a language barrier.
- Finding a safe location to travel to with low crime and corruption.
- Learning local transportation systems

FAVORITE BRANDS





"I'm making a more conscious effort to not scroll on my phone in bed. I think it takes me less time to fall asleep, and I feel more rested. The thing is that I like to read on my phone at night and have something to fall asleep to"

Lucy Chang

Age: 22

Location: Vancouver, BC

Education: Bachelors Degree in Comp. Sci.

Occupation: Front End Developer at AWS

Sleeping Habits:

- Goes to bed at around 12 a.m. everyday
- Sleeps at around 1:30 a.m. almost everyday
- Gets around 7 hours of sleep on average
- Takes 15 mins to fall asleep on average



Bio:

Lucy is a recent graduate working as a Front End Developer for AWS. Now out of school, Lucy's 8+ hour work days have placed new challenges on her sleeping habits. A self-proclaimed "night owl" during college, Lucy now wants to be able to sleep earlier to feel better throughout the day. She finds that without her phone in bed she feels bored, and unable to get rid of all her notifications.

Lucy is frustrated because it seems that no matter how many attempts she makes at sleeping early, she doesn't actually fall asleep until after 1 a.m. This has made her mornings feel rushed and unproductive, making her feel tired during the second half of the day. Recently she read an article which led her to cut out coffee, go for walks around Stanley Park, and activate "nightmode" and "do not disturb" in all her devices to sleep earlier.



Bedtime Behaviours:

- Spends a total of 30 to 90 minutes on her phone lying in bed before sleep
- Checks Gmail, WhatsApp, and Messenger to answer messages she couldn't get to during the day
- Tries to avoid mobile games because they increase her heart rate and make it harder to fall asleep
- Goes on Instagram and TikTok because the "scrolling" lets her pass the time easier
- Lowers screen brightness and turns on nightmode to watch YouTube or read on Kindle/Medium apps.



Personal Goals:

- Wants to go to sleep earlier to be more productive in the morning
- Wants to feel like she has more time and energy to do other activities outside of work during the day
- Avoid having to rely on coffee again as she did throughout her university years



Pain Points:

- Wishes that going to bed early would lead to falling asleep earlier, rather than past 1 a.m.
- Wishes she wouldn't feel bored in bed without her phone
- Wishes that she wouldn't have a dip in energy in the second half of her day



Apps:

Reading



Messaging



Entertainment



Example



Jim Krasinski

Age: Mid 20s

Single

Mobile developer

Hobbies: Sports, Music,
Games, Workout.

Needs & Goals

- A good way to spend spare time other than Netflix and video games
- Meet new people and make friends
- Develop professional knowledge to improve his career
- Learn new skills

Behaviours

- Too busy or lazy to research part-time course options
- Asks friends for advice
- Internet savvy with strong social media presence

"Routine life is so boring and doesn't allow me to progress. There are tons of things to learn but I hate doing research."



Martha White

Age: 50-55

Married with two kids

Housewife

Hobbies: Painting, Yoga,
Cooking.

Needs & Goals

- Spend time on meaningful things
- Meet young people and know latest trends
- Pick up past hobbies
- Live a healthy lifestyle

Behaviours

- Has lots of time to learn new skills
- Feels overwhelmed when doing google search
- Trusts other people's recommendations

"Now that all my kids are in college, I have more time to do my own things. But it's hard to find courses that suit me."

Requirements description

- Complex projects require the creation of a detailed document describing all the interaction requirements for the envisioned product

Name of major feature or category

Name of second-level feature or category

Requirement statement [WAAD source node ID]

Rationale (if useful): Rationale statement

Note (optional): Commentary about this requirement

Security

Privacy of ticket-buyer transactions

Shall protect security and privacy of ticket-buyer transactions [C19]

Note: In design, consider timeout feature to clear screen between customers.

Finding events

Direct keyword search by event description

Ticket buyers shall be able to find (e.g., search) by content to identify relevant current and future events [CA9].

Browse events by parameters

Ticket buyers shall be able to browse by category, description, location, time, rating, and price.

Feature list / Requirements brief

- In less complex projects, a list of the features that should be included in the product (with a short description) is sufficient

STEP 1 LIST FEATURES BEING CONSIDERED
Secure Sign In
Data Import from Outlook
Data Import from Social Networks
Data Import from OS Activity
Personalized Reporting Dashboard
Goals Manager
Suggested Activities & Content
Automatic Data Syncing
"Try it Out" Calendar
Mobile Reminders
Profile
Connections & Inbuilt Social Network
Behavior Models

Requirements (What)

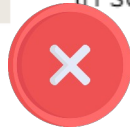
Socially networked. Integrates with other social networks. Easy for users to import data from other social networks, and easy for them to share what they've been doing in the consumer product with outside networks.

Device interoperable and mobile enabled. Designed for mobile first. Smartphone and SMS capabilities create a daily dialog with the user. PC and tablet experiences invite configuration and deeper integration with content.

Supports formal and informal goal setting. System suggests and detects possible goals, and also enables users to manually create their own goals.

Brings in data from a variety of places. This is the heart of the system. Integrates with tools like Outlook, iCal, Google Calendar, and other productivity software.

Rich information visualizations. Data is repackaged and displayed in suprising and engaging ways.

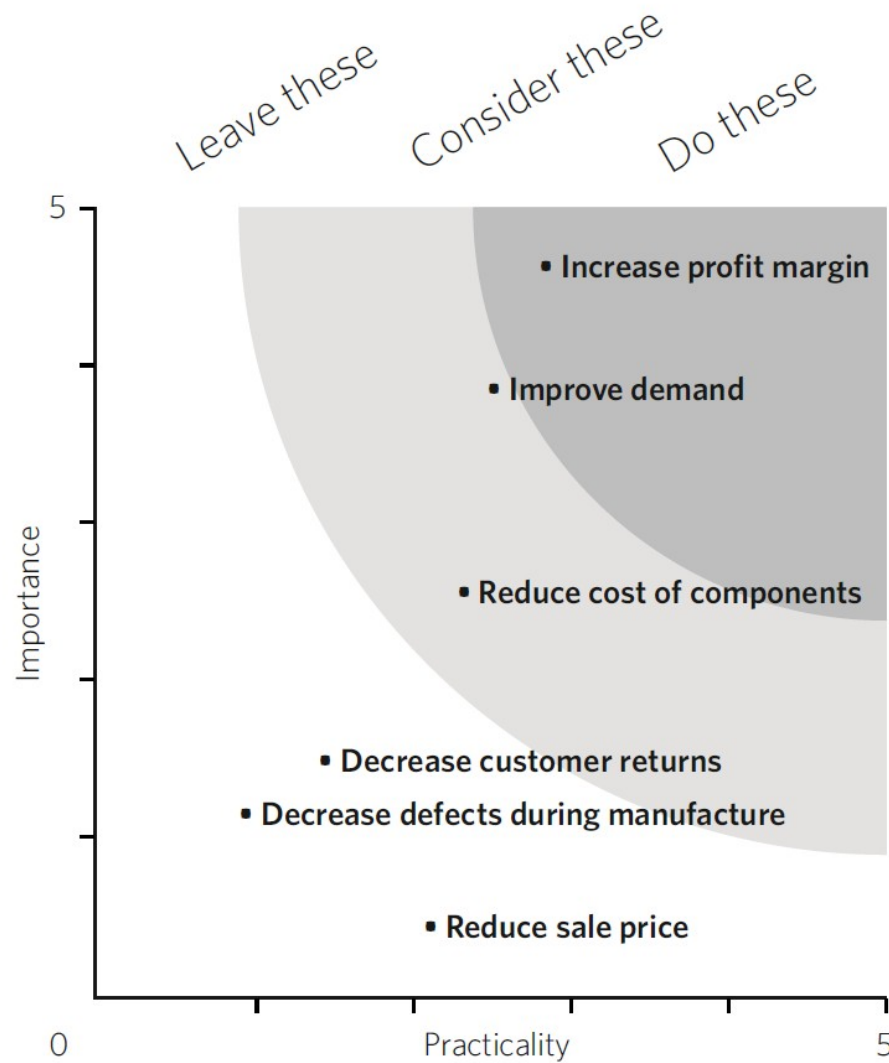


Requirements definition

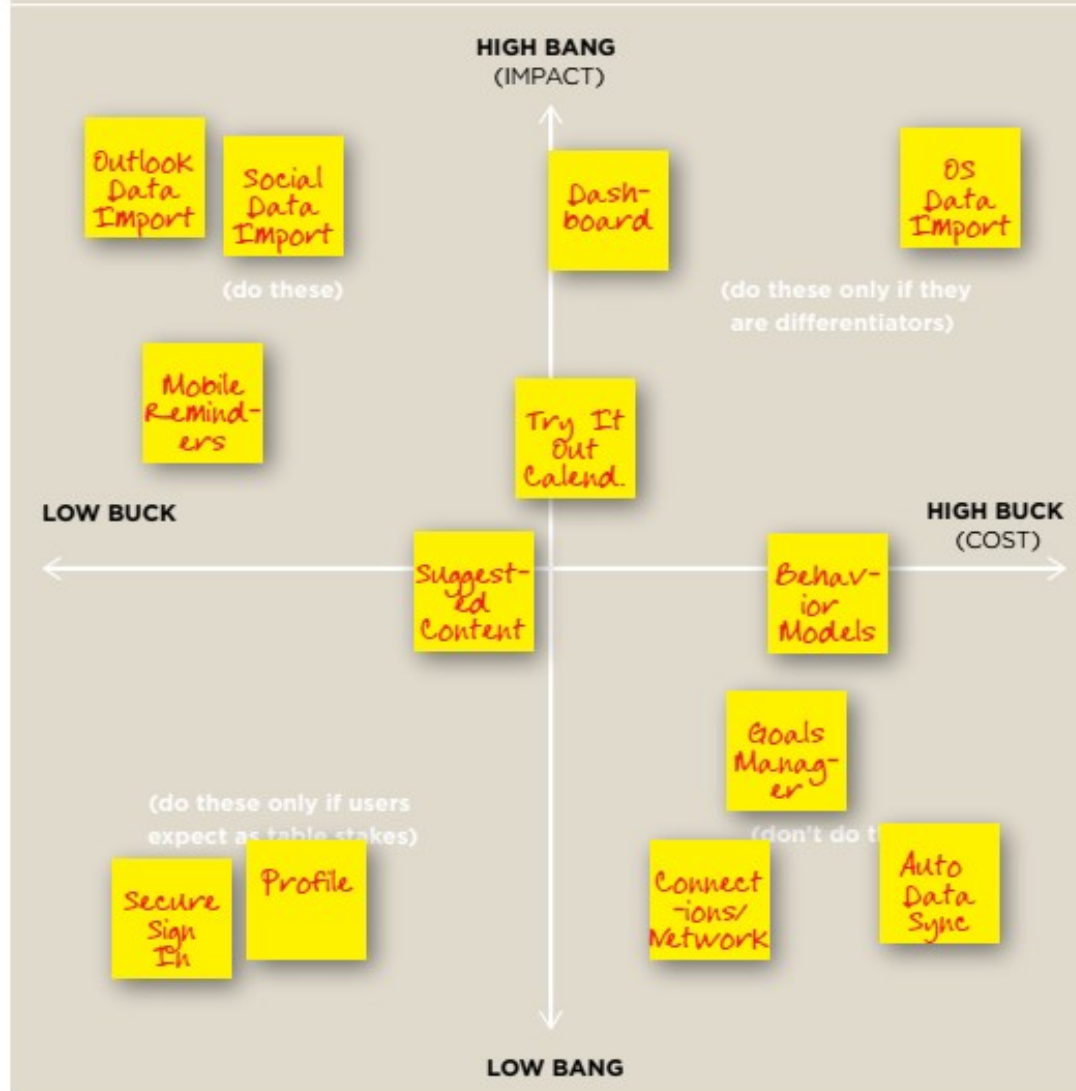
- The list of features must be complete and clear
- Focus on what features to include, not on how these should be implemented
- Always ask yourself “What is the user trying to do here?”
- Example: user registration
 - Is the user trying to register because they want to protect something they value (such as their bank details)? Is it because they want to get a reward you’re offering them (which is a weaker reason)? Is it because we’re making them do it, and it’s a barrier we’ve put in their way?

Requirements prioritization

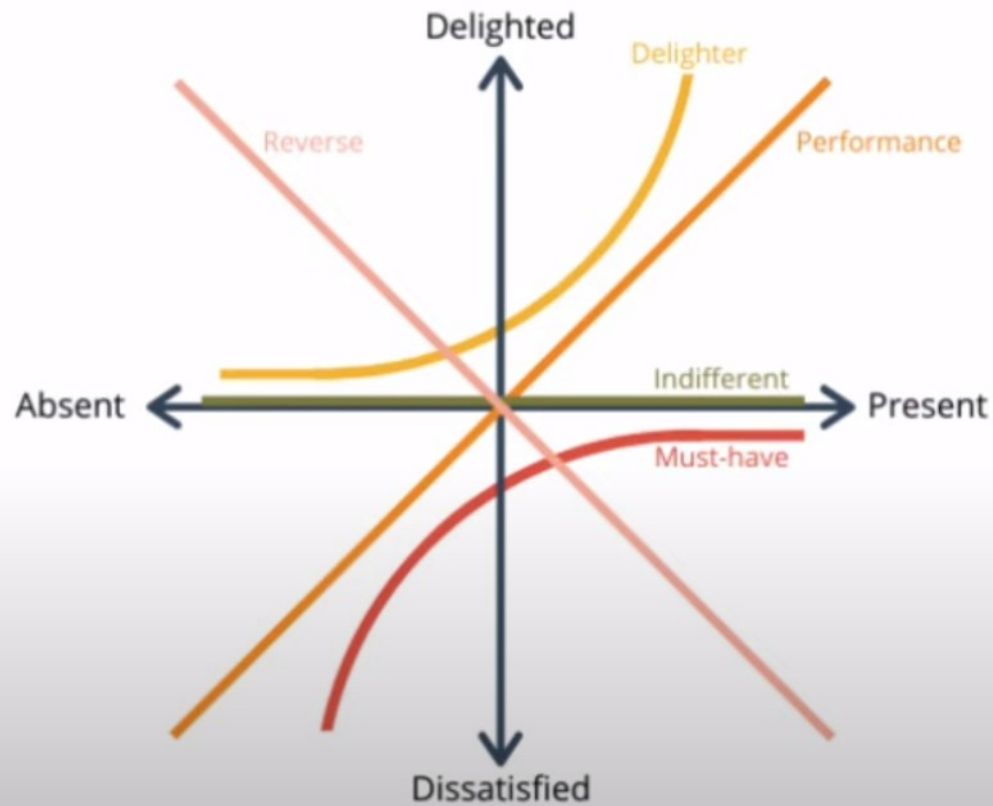
- Your resources are typically limited
- Once you have a list of requirements you must prioritize them
- You can use different prioritization models
 - The simplest is to create a qualitative prioritization based on levels (e.g., high/medium/low)
 - Involve users in defining the prioritization (e.g., by presenting pairs of requirements and asking them to choose the most important one)
 - More complex prioritization models involve multiple criteria



STEP 2 PLOT FEATURES ON BANG VS. BUCK CHART



The Kano Model



- Start with the must-haves.
- Then include as many performance features as you can.
- Next, add your delight features.
 - Cut performance features if necessary so you can include some.
- Remove any reverse features if you have any.
- Ignore indifferent features.